MATH 345 Skeleton Notes
Unit 4: Orthogonality and least squares
Section 4.2: Orthogonal projection

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We introduce orthogonal projection and then use it to derive the Gram–Schmidt algorithm.

Definition and formula

Definition: The orthogonal complement of a subspace

If U is a subspace of $\mathbb{R}^n$, the [ ] of U is the subset of $\mathbb{R}^n$ given by

$$U^\perp = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{y} = 0 \text{ for all } \mathbf{y} \text{ in } U\}$$

Theorem: Properties of orthogonal complements

The following properties of the orthogonal complement hold:

1. For any subspace U of $\mathbb{R}^n$, $U^\perp$ is a subspace of $\mathbb{R}^n$
2. $\{\mathbf{0}\}^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \{\mathbf{0}\}$
3. If $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, then

$$U^\perp = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{x}_i = 0 \text{ for } i = 1, 2, \dots, k\}$$

Activity: Finding an orthogonal complement (U4-LO2)

Find $U^{\perp}$ if $U = \text{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$.

Fact: Orthogonality after subtracting components

Let $\{\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_m\}$ be an [ ] in $\mathbb{R}^n$. Given $\mathbf{x}$ in $\mathbb{R}^n$, define

$$\mathbf{v} = \mathbf{x} - (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 - (\mathbf{x} \cdot \mathbf{q}_2)\mathbf{q}_2 - \cdots - (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m.$$

Then $\mathbf{v} \cdot \mathbf{q}_k = 0$ for $k = 1, 2, \ldots, m$. Equivalently, $\mathbf{v}$ lies in the orthogonal complement of the span:

$$\mathbf{v} \in (\text{span}\{\mathbf{q}_1, \ldots, \mathbf{q}_m\})^{\perp}.$$

Why is this true?

Let $c_i = \mathbf{x} \cdot \mathbf{q}_i$ for each i. Given $1 \leq k \leq m$, we calculate that

$$\begin{aligned}\mathbf{v} \cdot \mathbf{q}_k &= (\mathbf{x} - c_1\mathbf{q}_1 - c_2\mathbf{q}_2 - \cdots - c_m\mathbf{q}_m) \cdot \mathbf{q}_k \\ &= \mathbf{x} \cdot \mathbf{q}_k - c_1(\mathbf{q}_1 \cdot \mathbf{q}_k) - \cdots - c_k(\mathbf{q}_k \cdot \mathbf{q}_k) - \cdots - c_m(\mathbf{q}_m \cdot \mathbf{q}_k)\end{aligned}$$

Since the set is orthonormal, $\mathbf{q}_i \cdot \mathbf{q}_k = 0$ for $i \neq k$ and $\mathbf{q}_k \cdot \mathbf{q}_k = 1$. Thus

$$\begin{aligned}\mathbf{v} \cdot \mathbf{q}_k &= \mathbf{x} \cdot \mathbf{q}_k - c_k \\ &= \mathbf{x} \cdot \mathbf{q}_k - \mathbf{x} \cdot \mathbf{q}_k = 0\end{aligned}$$

Remark: When the vector is already in the span

If $\mathbf{x} \in \operatorname{span}\{\mathbf{q}_1, \ldots, \mathbf{q}_m\}$, the expansion theorem (Expansion theorem) gives

$$\mathbf{x} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m.$$

Thus the vector $\mathbf{v}$ in Orthogonality after subtracting components is $\mathbf{0}$.

Definition: The orthogonal projection onto a subspace

Let U be a subspace of $\mathbb{R}^n$. Then

the [ ] of the vector $\mathbf{x} \in \mathbb{R}^n$

onto U is the vector $\mathbf{y} = \operatorname{proj}_U(\mathbf{x}) \in U$ which is *closest* to $\mathbf{x}$, in the sense that for any $\mathbf{z} \in U$ with $\mathbf{z} \neq \mathbf{y}$, $\|\mathbf{x} - \mathbf{y}\| < \|\mathbf{x} - \mathbf{z}\|$.

The following theorem gives us a practical way to calculate orthogonal projections.

Theorem: Projection theorem

Let U be a subspace of $\mathbb{R}^n$ with [ ]
$\{\mathbf{q}_1, \ldots, \mathbf{q}_m\}$. For any vector $\mathbf{x} \in \mathbb{R}^n$, define

$$\mathbf{p} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{x} \cdot \mathbf{q}_2)\mathbf{q}_2 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m.$$

Then $\mathbf{p} = \text{proj}_U(\mathbf{x})$. The difference

$$\mathbf{v} = \mathbf{x} - \mathbf{p} = \mathbf{x} - \text{proj}_U(\mathbf{x})$$

is called the [ ] and lies in $U^\perp$. In particular, if $\mathbf{x}$ is already in U, then $\mathbf{x}$ equals $\mathbf{p}$.

Why is this true?

If $\{\mathbf{q}_1, \ldots, \mathbf{q}_m\}$ is an [ ] for U, and we define

$$\mathbf{p} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{x} \cdot \mathbf{q}_2)\mathbf{q}_2 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m,$$

then $\mathbf{p}$ is a linear combination of the basis vectors, so $\mathbf{p} \in U$. By Orthogonality after subtracting components, $\mathbf{x} - \mathbf{p}$ is orthogonal to every basis vector, so $\mathbf{x} - \mathbf{p} \in U^\perp$. Thus if $\mathbf{z} \in U$ and $\mathbf{z} \neq \mathbf{p}$, the Pythagorean theorem (Pythagoras' theorem) tells us that

$$\begin{aligned} \|\mathbf{x} - \mathbf{z}\|^2 &= \|(\mathbf{x} - \mathbf{p}) + (\mathbf{p} - \mathbf{z})\|^2 \\ &= \|\mathbf{x} - \mathbf{p}\|^2 + \|\mathbf{p} - \mathbf{z}\|^2. \end{aligned}$$

Since $\mathbf{p} \neq \mathbf{z}$, we find that

$$\|\mathbf{x} - \mathbf{z}\|^2 > \|\mathbf{x} - \mathbf{p}\|^2,$$

and taking square roots gives that

$$\|\mathbf{x} - \mathbf{z}\| > \|\mathbf{x} - \mathbf{p}\|.$$

Thus the vector $\mathbf{p}$ we have defined is equal to $\text{proj}_U(\mathbf{x})$. Since we have already verified that $\mathbf{x} - \mathbf{p} = \mathbf{x} - \text{proj}_U(\mathbf{x})$ is in $U^\perp$, the theorem follows.

Video walkthrough of Projection theorem: Watch: The projection theorem (1:15).

Orthogonal projection decomposes the original vector into a vector in the subspace and a residual perpendicular to the subspace.

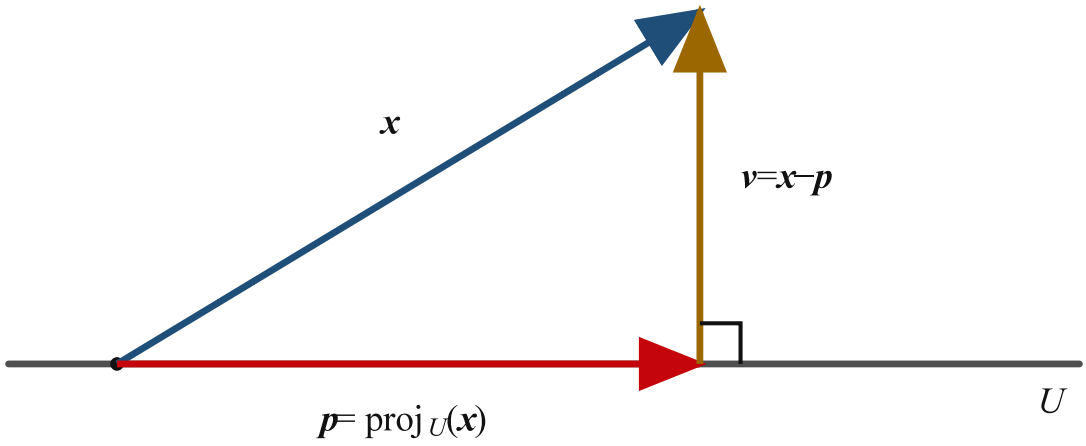


Figure. Projection and residual. Orthogonal projection writes $\mathbf{x} = \mathbf{p} + \mathbf{v}$, where $\mathbf{p} \in U$ and $\mathbf{v} \in U^{\perp}$.

Warning: Normalize before using the projection formula

The formula in Projection theorem requires an *orthonormal* basis: mutual orthogonality alone is not enough. If you are given only an orthogonal set $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ spanning U, first normalize each vector:

$$\mathbf{q}_i = \frac{\mathbf{f}_i}{\|\mathbf{f}_i\|}, \qquad i = 1, \dots, m.$$

The vectors $\mathbf{q}_i$ form an orthonormal basis of the same subspace. Use these unit vectors in the projection formula.

Note: Expansion versus projection

The expansion theorem and the projection theorem use the same orthonormal basis and the same coefficient formula. The difference in their hypotheses is that the expansion theorem requires $\mathbf{x} \in U = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_m\}$, so the sum equals $\mathbf{x}$ itself. The projection theorem allows any $\mathbf{x} \in \mathbb{R}^n$; the sum equals $\operatorname{proj}_U(\mathbf{x})$, which need not

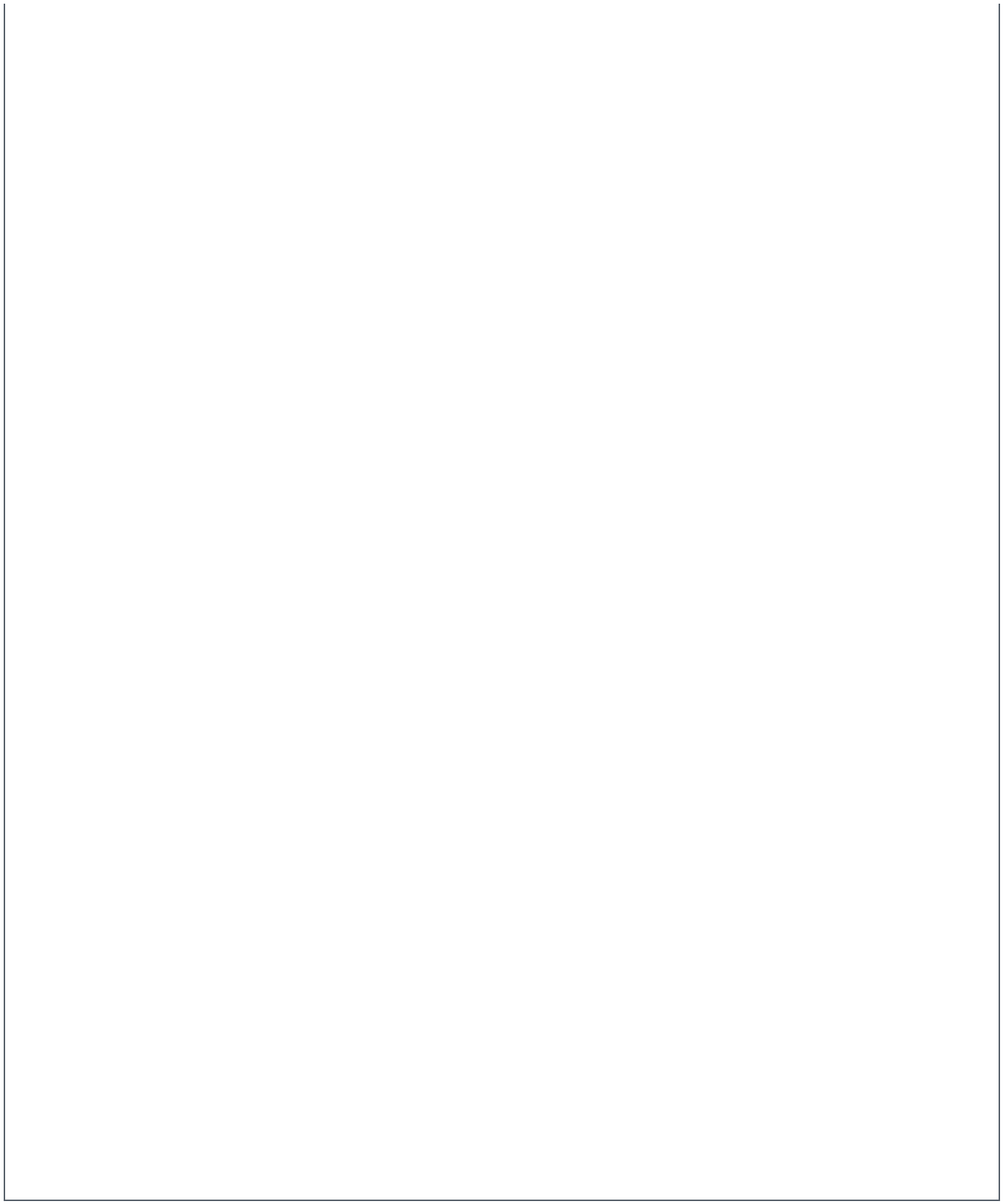
equal $\mathbf{x}$. When $\mathbf{x} \in U$, the projection equals $\mathbf{x}$ and the residual is zero, recovering the expansion theorem.

Example: Projecting onto a span (U4-LO3)

Find $\text{proj}_U \mathbf{x}$ where $U = \text{span}\{(1, 1, 0), (1, -1, 0)\}$ and $\mathbf{x} = (2, 0, 1)$ in $\mathbb{R}^3$.

Activity: Projection onto a plane (U4-LO3)

Find $\text{proj}_U \mathbf{x}$ where $U = \text{span}\{(1, 0, 1), (1, 0, -1)\}$ and $\mathbf{x} = (1, 2, 1)$ in $\mathbb{R}^3$.



Activity: Projection onto a line (U4-LO3)

Let

$$\mathbf{f} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \qquad L = \operatorname{span}\{\mathbf{f}\}.$$

1. Compute $\operatorname{proj}_L(\mathbf{x})$.
2. Compute the residual $\mathbf{v} = \mathbf{x} - \operatorname{proj}_L(\mathbf{x})$.
3. Check that $\mathbf{v} \cdot \mathbf{f} = 0$.

Activity: Projection onto a line in code (U4-LO3, U4-LO7)

The following code first normalizes the direction vector, then computes the same projection and residual as the previous activity.

```
import numpy as np

f = np.array([3.0, 4.0])
x = np.array([2.0, 5.0])

q = f / np.linalg.norm(f)
p = (x @ q) * q
v = x - p

p, v, v @ f
```

1. Which variable stores the projection?
2. Which variable stores the residual?
3. Why should `v @ f` be close to zero?
4. What does `q = f / np.linalg.norm(f)` accomplish, and why is it needed before computing `(x @ q) * q`?

Projection as an optimization problem

The closest-point definition of projection naturally leads us to consider projection as an optimization problem.

Activity: Projection as optimization (U4-LO3, U3-LO9)

Let

$$W = \operatorname{span}\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\} \subseteq \mathbb{R}^2, \qquad \mathbf{b} = \begin{bmatrix}2\\3\end{bmatrix}.$$

A general point of W has the form

Activity: Projection as optimization (U4-LO3, U3-LO9) (continued)

$$\mathbf{w}(t) = \begin{bmatrix} t \\ 0 \end{bmatrix}.$$

1. Write the squared distance $g(t) = \|\mathbf{b} - \mathbf{w}(t)\|^2$.
2. Compute $g'(t)$ and find the critical point of g.
3. Use $g''(t)$ to classify the critical point. (Recall the second derivative test: if $g'(t_0) = 0$ and g'' is continuous near t_0, then $g''(t_0) > 0$ gives a local minimum, $g''(t_0) < 0$ gives a local maximum, and $g''(t_0) = 0$ is inconclusive; see OpenStax Calculus Volume 1, Section 4.5.)
4. What point of W is closest to $\mathbf{b}$?
5. How does this agree with the figure and Projection theorem?

Example: Deriving the projection formula through critical points

Let $\mathbf{x} \in \mathbb{R}^n$, and let $B = (\mathbf{q}_1, \dots, \mathbf{q}_m)$ be an ordered orthonormal basis of U, so

$$U = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_m\}.$$

Every vector in U can be written uniquely as

$$\mathbf{u}(\mathbf{c}) = c_1\mathbf{q}_1 + \cdots + c_m\mathbf{q}_m, \qquad \mathbf{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix} \in \mathbb{R}^m.$$

Recall from Coordinate Vectors With Respect to a Basis that $\mathbf{c} = C_B(\mathbf{u}(\mathbf{c}))$ is the coordinate vector of $\mathbf{u}(\mathbf{c})$ with respect to the ordered basis B. Finding the closest vector to $\mathbf{x}$ is therefore equivalent to choosing $\mathbf{c}$ to minimize

$$g(\mathbf{c}) = \|\mathbf{x} - \mathbf{u}(\mathbf{c})\|^2.$$

Squaring the distance does not change which vector is closest, and it gives a polynomial that is easy to differentiate.

First use the definition of the squared norm and distribute the dot product:

$$\begin{aligned} g(\mathbf{c}) &= (\mathbf{x} - \mathbf{u}(\mathbf{c})) \cdot (\mathbf{x} - \mathbf{u}(\mathbf{c})) \\ &= (\mathbf{x} - c_1\mathbf{q}_1 - \cdots - c_m\mathbf{q}_m) \\ &\qquad \cdot (\mathbf{x} - c_1\mathbf{q}_1 - \cdots - c_m\mathbf{q}_m) \\ &= \mathbf{x} \cdot \mathbf{x} - \mathbf{x} \cdot \mathbf{u}(\mathbf{c}) \\ &\qquad - \mathbf{u}(\mathbf{c}) \cdot \mathbf{x} + \mathbf{u}(\mathbf{c}) \cdot \mathbf{u}(\mathbf{c}) \\ &= \|\mathbf{x}\|^2 - 2\mathbf{x} \cdot \mathbf{u}(\mathbf{c}) + \mathbf{u}(\mathbf{c}) \cdot \mathbf{u}(\mathbf{c}). \end{aligned}$$

The last equality uses symmetry of the dot product: $\mathbf{u}(\mathbf{c}) \cdot \mathbf{x} = \mathbf{x} \cdot \mathbf{u}(\mathbf{c})$. To expand the middle term, substitute the linear combination for $\mathbf{u}(\mathbf{c})$, distribute, and pull out the scalar coefficients:

$$\begin{aligned}\mathbf{x}\cdot\mathbf{u}(\mathbf{c}) &= \mathbf{x}\cdot(c_1\mathbf{q}_1+\cdots+c_m\mathbf{q}_m)\\ &= c_1(\mathbf{x}\cdot\mathbf{q}_1)+\cdots+c_m(\mathbf{x}\cdot\mathbf{q}_m).\end{aligned}$$

For the last term, first compute the dot product of each basis vector with $\mathbf{u}(\mathbf{c})$. For any $i = 1, \dots, m$,

$$\begin{aligned}\mathbf{q}_i\cdot\mathbf{u}(\mathbf{c}) &= \mathbf{q}_i\cdot(c_1\mathbf{q}_1+\cdots+c_m\mathbf{q}_m)\\ &= c_1(\mathbf{q}_i\cdot\mathbf{q}_1)+\cdots+c_m(\mathbf{q}_i\cdot\mathbf{q}_m).\end{aligned}$$

Here is where orthonormality is used. Orthogonality gives $\mathbf{q}_i\cdot\mathbf{q}_j = 0$ whenever $j \neq i$, so every term except the ith is zero. The unit-length condition gives $\mathbf{q}_i\cdot\mathbf{q}_i = \|\mathbf{q}_i\|^2 = 1$. Therefore

$$\mathbf{q}_i\cdot\mathbf{u}(\mathbf{c}) = c_i(\mathbf{q}_i\cdot\mathbf{q}_i) = c_i\cdot 1 = c_i.$$

Distributing once more now gives

$$\begin{aligned}\mathbf{u}(\mathbf{c})\cdot\mathbf{u}(\mathbf{c}) &= (c_1\mathbf{q}_1+\cdots+c_m\mathbf{q}_m)\cdot\mathbf{u}(\mathbf{c})\\ &= c_1(\mathbf{q}_1\cdot\mathbf{u}(\mathbf{c}))+\cdots+c_m(\mathbf{q}_m\cdot\mathbf{u}(\mathbf{c}))\\ &= c_1c_1+\cdots+c_mc_m\\ &= c_1^2+\cdots+c_m^2.\end{aligned}$$

Thus orthogonality removes the cross terms between distinct basis vectors, and normalization makes the coefficient of each c_i^2 equal to one. Substituting both dot-product calculations into the formula for g, we obtain

$$\begin{aligned}g(\mathbf{c}) &= \|\mathbf{x}\|^2 - 2\big[c_1(\mathbf{x}\cdot\mathbf{q}_1)+\cdots+c_m(\mathbf{x}\cdot\mathbf{q}_m)\big]\\ &\quad + c_1^2+\cdots+c_m^2\\ &= \|\mathbf{x}\|^2 + \big[c_1^2 - 2c_1(\mathbf{x}\cdot\mathbf{q}_1)\big]\\ &\quad +\cdots+\big[c_m^2 - 2c_m(\mathbf{x}\cdot\mathbf{q}_m)\big].\end{aligned}$$

Now fix i and differentiate with respect to c_i. The vector $\mathbf{x}$ and all basis vectors are fixed, so their dot products are constants. In a partial derivative, the other coordinates c_j, for $j \neq i$, are also held constant. Hence

$$\frac{\partial}{\partial c_i}\|\mathbf{x}\|^2 = 0,$$

and

$$\frac{\partial}{\partial c_i}\left[c_j^2 - 2c_j(\mathbf{x}\cdot\mathbf{q}_j)\right] = 0 \qquad \text{if } j \neq i.$$

Only the term involving c_i contributes. By linearity of differentiation and the power rule,

$$\begin{aligned}\frac{\partial g}{\partial c_i} &= \frac{\partial}{\partial c_i}\left[c_i^2 - 2c_i(\mathbf{x}\cdot\mathbf{q}_i)\right] \\ &= \frac{\partial c_i^2}{\partial c_i} - 2(\mathbf{x}\cdot\mathbf{q}_i)\frac{\partial c_i}{\partial c_i} \\ &= 2c_i - 2(\mathbf{x}\cdot\mathbf{q}_i)\cdot 1 \\ &= 2c_i - 2(\mathbf{x}\cdot\mathbf{q}_i).\end{aligned}$$

This holds for each $i = 1, \ldots, m$. Setting them all equal to zero gives the unique critical point $\mathbf{c}^*$, whose entries are

$$c_i^* = \mathbf{x}\cdot\mathbf{q}_i, \qquad i = 1, \ldots, m.$$

To see that this critical point is the unique global minimum, "complete the squares". For each i, add and subtract $(c_i^*)^2$:

$$\begin{aligned}c_i^2 - 2c_ic_i^* &= c_i^2 - 2c_ic_i^* + (c_i^*)^2 - (c_i^*)^2 \\ &= (c_i - c_i^*)^2 - (c_i^*)^2.\end{aligned}$$

Since $c_i^* = \mathbf{x}\cdot\mathbf{q}_i$, substituting this identity into the expanded formula for g gives

$$g(\mathbf{c}) = \|\mathbf{x}\|^2 - \left[(c_1^*)^2 + \cdots + (c_m^*)^2\right] + (c_1 - c_1^*)^2 + \cdots + (c_m - c_m^*)^2.$$

At $\mathbf{c} = \mathbf{c}^*$, all the squared differences are zero, so

$$g(\mathbf{c}^*) = \|\mathbf{x}\|^2 - \left[(c_1^*)^2 + \cdots + (c_m^*)^2\right].$$

Therefore

$$g(\mathbf{c}) = g(\mathbf{c}^*) + (c_1 - c_1^*)^2 + \cdots + (c_m - c_m^*)^2.$$

The added terms are nonnegative and all vanish exactly when $\mathbf{c} = \mathbf{c}^*$. Consequently, the closest vector in U is

$$\begin{aligned}\mathbf{p} = \mathbf{u}(\mathbf{c}^*) &= c_1^*\mathbf{q}_1 + \cdots + c_m^*\mathbf{q}_m \\ &= (\mathbf{x}\cdot\mathbf{q}_1)\mathbf{q}_1 + \cdots + (\mathbf{x}\cdot\mathbf{q}_m)\mathbf{q}_m \\ &= \operatorname{proj}_U(\mathbf{x}).\end{aligned}$$

This recovers the formula in Projection theorem by minimizing squared distance.

Directional derivatives as projection coefficients

Let $\mathbf{v}$ be a unit vector. By Projection theorem,

$$\operatorname{proj}_{\operatorname{span}\{\mathbf{v}\}}\big(\nabla f(\mathbf{a})\big) = \big(\nabla f(\mathbf{a})\cdot\mathbf{v}\big)\mathbf{v}.$$

By Directional derivatives from the gradient,

$$\operatorname{proj}_{\operatorname{span}\{\mathbf{v}\}}\big(\nabla f(\mathbf{a})\big) = D_{\mathbf{v}}f(\mathbf{a})\mathbf{v}.$$

Thus $D_{\mathbf{v}}f(\mathbf{a})$ is the signed projection coefficient of the gradient in the oriented direction $\mathbf{v}$. Its absolute value is the length of the projection.

Activity: Directional derivative as a projection coefficient (U4-LO3, U3-LO8)

Recall the gradient and unit vectors from Alignment with the gradient (U3-LO8, U1-LO1):

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix},$$

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$
$$\mathbf{v}_3 = \frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \qquad \mathbf{v}_4 = -\frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

For each unit vector $\mathbf{v}_i$, compute

$$\text{proj}_{\text{span}\{\mathbf{v}_i\}}\big(\nabla f(\mathbf{a})\big) = D_{\mathbf{v}_i} f(\mathbf{a})\mathbf{v}_i.$$

1. Which projection coefficients are positive, and which are negative?
2. Why do $\mathbf{v}_3$ and $\mathbf{v}_4$ give the same projection vector but opposite directional derivatives?

Gram-Schmidt orthonormalization

The Gram-Schmidt algorithm converts a basis into an orthonormal basis of the same subspace through a sequence of projection, subtraction, measurement, and normalization steps.

Fact: Expanding an orthonormal set

Let $\{\mathbf{q}_1, \dots, \mathbf{q}_m\}$ be an [ ] in $\mathbb{R}^n$, and let $\mathbf{x} \in \mathbb{R}^n$ lie [ ]. Define

$$\mathbf{v} = \mathbf{x} - (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 - \cdots - (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m.$$

Then $\mathbf{v} \neq \mathbf{0}$. [ ] it by setting

$$\mathbf{q}_{m+1} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

The enlarged set $\{\mathbf{q}_1, \dots, \mathbf{q}_m, \mathbf{q}_{m+1}\}$ is an [ ], and the span is preserved:

$$\operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_m, \mathbf{q}_{m+1}\} = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_m, \mathbf{x}\}.$$

Why is this true?

If $\mathbf{v} = \mathbf{0}$, then $\mathbf{x} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m$ lies in the original span, contrary to the hypothesis. Thus $\mathbf{v} \neq \mathbf{0}$ and we can normalize it. By Orthogonality after subtracting components, $\mathbf{v}$ is orthogonal to every $\mathbf{q}_k$. The vector $\mathbf{q}_{m+1}$ has norm one and remains orthogonal to every $\mathbf{q}_k$, so the enlarged set is orthonormal.
The formula for $\mathbf{q}_{m+1}$ expresses it as a linear combination of $\mathbf{q}_1, \dots, \mathbf{q}_m, \mathbf{x}$. Conversely,

$$\mathbf{x} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m + \|\mathbf{v}\|\mathbf{q}_{m+1}.$$

Each set therefore spans the other, proving the equality of spans.

Activity: Extending an orthonormal set to a basis (U4-LO2)

First normalize the orthogonal vectors

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

to obtain an orthonormal set $\{\mathbf{q}_1, \mathbf{q}_2\}$ in $\mathbb{R}^3$. Use Expanding an orthonormal set with $\mathbf{x} = (1, 1, 2)$ to extend this set to an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$.

We use an original column $\mathbf{a}_j$, its projection $\mathbf{p}_j$, the orthogonal residual $\mathbf{v}_j$, the new unit direction $\mathbf{q}_j$, and coefficients r_{ij}. Repeatedly applying Expanding an orthonormal set to the normalized vectors $\mathbf{q}_i$ gives the algorithm below.

Algorithm: Gram-Schmidt algorithm

A matrix

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$$

with linearly independent columns.
For $j = 1, \ldots, n$:

1. Start with $\mathbf{a}_j$.
2. For each $i < j$, compute

$$r_{ij} = \mathbf{q}_i^T \mathbf{a}_j.$$

   Then assemble

$$\mathbf{p}_j = \sum_{i<j} r_{ij} \mathbf{q}_i.$$

   For the first column, $\mathbf{p}_1 = \mathbf{0}$.

3.

$$\mathbf{v}_j = \mathbf{a}_j - \mathbf{p}_j.$$

4. [ ]

$$r_{jj} = \|\mathbf{v}_j\|.$$

5. [ ]

$$\mathbf{q}_j = \frac{\mathbf{v}_j}{r_{jj}}.$$

[ ] The vectors $\mathbf{q}_1, \ldots, \mathbf{q}_n$ are orthonormal, and

$$\operatorname{span}\{\mathbf{q}_1, \ldots, \mathbf{q}_j\} = \operatorname{span}\{\mathbf{a}_1, \ldots, \mathbf{a}_j\}$$

at every stage.

Practice Gram-Schmidt geometrically in Perfectly Normal.

Warning

If $r_{jj} = 0$, then $\mathbf{a}_j$ is already in the span of the earlier columns. The input columns are not linearly independent, and no new unit direction is produced.

The following theorem summarizes the properties of the vectors produced by Gram-Schmidt algorithm.

Theorem: Properties of the Gram-Schmidt output

Apply Gram-Schmidt algorithm to a matrix $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ with linearly independent columns, and let $U = \operatorname{span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$. For each $k = 1, \ldots, n$, write $U_k = \operatorname{span}\{\mathbf{a}_1, \ldots, \mathbf{a}_k\}$. The output through step k satisfies:

- The span is preserved:

$$\operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\} = U_k.$$

- The normalized vectors $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ form an orthonormal basis of U_k.

In particular, $\{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ is an orthonormal basis of U.

Example: Gram–Schmidt for a column space (U4-LO2)
Apply the Gram-Schmidt algorithm to the columns of A, where

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}.$$

At each stage, record the projection coefficients, projection, residual, residual length, and normalized direction. Give an orthogonal basis and an orthonormal basis for $\operatorname{col}(A)$ (recall Column and Row Space).

1. Compute $\mathbf{v}_1$, r_{11}, and $\mathbf{q}_1$.
2. Compute r_{12}, $\mathbf{p}_2$, $\mathbf{v}_2$, r_{22}, and $\mathbf{q}_2$.
3. Compute r_{13}, r_{23}, $\mathbf{p}_3$, $\mathbf{v}_3$, r_{33}, and $\mathbf{q}_3$.
4. Give an orthogonal basis for $\operatorname{col}(A)$ from the residuals.
5. Give an orthonormal basis for $\operatorname{col}(A)$ from the unit vectors.
6. Explain why both bases span $\operatorname{col}(A)$.

Activity: The closest point on a plane (U4-LO3)

Find the point in the plane defined by the equation $x - y - 2z = 0$, which is closest to the point $(1, 2, -1)$.

